

# Apply filters to SQL queries

## Project description

In this project, I used SQL filters to investigate potential security issues and retrieve specific employee information from a database. I applied various SQL operators including AND, OR, NOT, and LIKE to filter data from the log\_in\_attempts and employees tables.

## Retrieve after hours failed login attempts

```
clear
MariaDB [organization]> SELECT *
  ->
  -> FROM log_in_attempts
  ->
  -> WHERE login_time > '18:00' AND success = 0;
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  | 0       |
| 20       | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  | 0       |
| 28       | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 34       | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   | 0       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 52       | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   | 0       |
| 69       | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  | 0       |
| 82       | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  | 0       |
| 87       | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 | 0       |
| 96       | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  | 0       |
| 104      | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  | 0       |
| 107      | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 | 0       |
| 111      | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   | 0       |
| 127      | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  | 0       |
| 131      | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 | 0       |
| 155      | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 | 0       |
| 160      | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  | 0       |
| 199      | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  | 0       |

```
19 rows in set (0.045 sec)

MariaDB [organization]>
```

I queried the log\_in\_attempts table to find all failed login attempts that occurred after business hours. I used the AND operator to filter for login attempts where the login\_time was after 18:00 and the success column equaled 0, indicating a failed attempt. This returned 19 rows.

## Retrieve login attempts on specific dates

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 24       | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 | 1       |
| 25       | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26       | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 28       | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 30       | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  | 1       |
| 32       | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 36       | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 38       | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 39       | yappiah  | 2022-05-09 | 07:56:40   | MEXICO  | 192.168.57.115  | 1       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43       | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |

75 rows in set (0.001 sec)

```
MariaDB [organization]>
```

I used the OR operator to retrieve all login attempts that occurred on 2022-05-09 or 2022-05-08. This allowed me to investigate a suspicious event by reviewing all login activity across both days.

## Retrieve login attempts outside of Mexico

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE NOT country LIKE 'MEX%';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 5        | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  | 0       |
| 7        | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 | 1       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 10       | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 | 0       |
| 11       | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 13       | mrhah    | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 | 1       |

```
144 rows in set (0.032 sec)
```

```
MariaDB [organization]>
```

I used the NOT operator combined with LIKE and the pattern 'MEX%' to retrieve all login attempts that did not originate in Mexico. Using LIKE with % ensured both 'MEX' and 'MEXICO' were excluded from the results.

## Retrieve employees in Marketing

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

| employee_id | device_id    | username | department | office   |
|-------------|--------------|----------|------------|----------|
| 1000        | a320b137c219 | elarson  | Marketing  | East-170 |
| 1052        | a192b174c940 | jdarosa  | Marketing  | East-195 |
| 1075        | x573y883z772 | fbautist | Marketing  | East-267 |
| 1088        | k865l965m233 | rgosh    | Marketing  | East-157 |
| 1103        | NULL         | randerss | Marketing  | East-460 |
| 1156        | a184b775c707 | dellery  | Marketing  | East-417 |
| 1163        | h679i515j339 | cwilliam | Marketing  | East-216 |

```
7 rows in set (0.001 sec)
```

I queried the employees table using AND to filter for employees in the Marketing department who are located in the East building. I used LIKE with the pattern 'East%' to capture all East building offices.

## Retrieve employees in Finance or Sales

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE department = 'Finance' OR department = 'Sales';
```

| employee_id | device_id    | username | department | office    |
|-------------|--------------|----------|------------|-----------|
| 1003        | d394e816f943 | sgilmore | Finance    | South-153 |
| 1007        | h174i497j413 | wjaffrey | Finance    | North-406 |
| 1008        | i858j583k571 | abernard | Finance    | South-170 |
| 1009        | NULL         | lrodriqu | Sales      | South-134 |
| 1010        | k242l212m542 | jlansky  | Finance    | South-109 |
| 1011        | l748m120n401 | drosas   | Sales      | South-292 |
| 1015        | p611q262r945 | jsoto    | Finance    | North-271 |
| 1017        | r550s824t220 | islark   | Finance    | North-188 |

```
71 rows in set (0.001 sec)  
  
MariaDB [organization]>
```

I used the OR operator to retrieve all employees in either the Finance or Sales department. Both conditions referenced the department column separately to return the complete list of employees from both departments.

## Retrieve all employees not in IT

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE NOT department = 'Information Technology';
```

| employee_id | device_id    | username | department      | office      |
|-------------|--------------|----------|-----------------|-------------|
| 1000        | a320b137c219 | elarson  | Marketing       | East-170    |
| 1001        | b239c825d303 | bmoreno  | Marketing       | Central-276 |
| 1002        | c116d593e558 | tshah    | Human Resources | North-434   |
| 1003        | d394e816f943 | sgilmore | Finance         | South-153   |
| 1004        | e218f877g788 | eraab    | Human Resources | South-127   |
| 1005        | f551g340h864 | gesparza | Human Resources | South-366   |

```
161 rows in set (0.001 sec)  
  
MariaDB [organization]>
```

I used the NOT operator to filter out employees in the Information Technology department, returning all other employees who still needed the security update applied to their machines.

## Summary

I used SQL filters to investigate security issues and retrieve employee information from the organization's database. Using AND, OR, NOT, and LIKE operators, I was able to narrow down login attempts by time, date, and location, and identify specific employee groups by department and office location. These queries helped support the security team in investigating suspicious activity and planning system updates.